UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	ICT450
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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QUESTION 1 (20 marks)

a) The file system method of organizing and managing data was a definite improvement over the manual system. Nonetheless, many problems and limitations become evident to this approach. Describe TWO (2) problems associated with a file system.

(4 marks)

Table : TRUCK

Truck_Num	Base_Code	Type_Code	Truck_Miles	Truck_Serial_Num
1001	501	1	5676.89	AA-322-1222-W11
1002	502	1	12232.67	AV-678-4545-Q23
1003	501	2	9886.68	FR-456-1123-R45
1004	501	2	23445.76	AD-213-7654-WQ
1005	502	3	454.98	XV-987-1234-SQ

Table : BASE

Base_Code	Base_City	Base_State	Base_Postcode	Base_Manager
501	Jasin	Malacca	77000	Fattah Aziz
502	Shah Alam	Selangor	40000	Michael Faraday
503	Kuala Kangsar	Perak	33000	Benjamin

Table : TYPE

Type_Code	Type_Description
1	Single box, double-axie
2	Single box, single axie
3	Tandem trailer, single axie

b) The Entity Relationship (ER) Model introduced by Chen in 1976 is widely accepted and adapted graphical tool for data modeling. Describe the basic structure of ER Model and discuss TWO (2) advantages of having an ER Model in database management system.

(4 marks)

c) Given the following tables :

i) Describe the type(s) of relationship(s) between TRUCK and BASE relations. Identify the entity integrity (if any) for each table. Answer YES or NO, and then explain your answer.

(5 marks)

ii) Draw the ER Diagram that corresponds to the above tables. Include all the appropriate connectivity and cardinalities of the relationships.

(7 marks)

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QUESTION 2 **(20 marks)**

a) A good database design uses appropriate data models as its foundation. Data model also has been classified based on degrees of data abstractions by American National Standards Institute (ANSI) and Standards Planning and Requirements Committee (SPARC).

i) Give ONE (1) importance of data model to database system environment.

(2 marks)

ii) Explain with illustration each level of data abstraction.

(8 marks)

b) Database design may be based on two different database design strategies: centralized and decentralized. Discuss an appropriate situation for each the design strategy.

(4 marks)

c) State stages of Database Life Cycle (DBLC) in sequence.

(6 marks)

QUESTION 3 (20 marks)

Table 1 shows a daily order for a new start-up company, Ridz's Burger. This order is manually recorded in a logbook. Each customer can make several orders and each order is managed by one staff.

Indication
B1 - Benjo
CK1 – Chicken Burger
CK2 – Chicken Cheese Burger
BF1 - Beef Burger
BF2 – Beef Cheese Burger
BF3 – Beef Special Burger

Based on the information given, answer the following questions:

Table 1: Ridz's Burger Logbook's Record Order

Date	Name of Customer	Total Orders	List of orders and Quantity	Total payment	Type of Payment	Name of Staff
1/2/2025	Intan	2	B1 - 1 CK2 - 1	RM 6.50	QR Code	Ridz
1/2/2025	Azwani	1	CK1 - 1	RM 4.00	Cash	Ridz
2/2/2025	Lya	1	BF3 - 1	RM 5.00	Cash	Syafinaz
4/2/2025	Lutfi	3	B1 - 3	RM 6.00	QR Code	Syafinaz

a) State THREE (3) problems that can be happened to the data in this logbook's record.

(3 marks)

b) Based on the scenario in Table 1, create the primary key(s) and identify partial dependencies and/or transitive dependencies.

(6 marks)

c) Normalize this logbook's record from the First Normal Form (1NF) until the Third Normal Form (3NF). List all the steps involved.

(11 marks)

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QUESTION 4 (20 marks)

Table 4 shows a relational schema for XY Company, a consulting company that tracks all assigned projects.

Table 4 : Relational Schema for XY Company

```
EMPLOYEE (EmpID, EmpFirstName, EmpLastName, EmpHireDate, JobCode)
JOB (JobCode, JobName, JobChargeHour)
PROJECT (ProjCode, ProjName)
ASSIGN (AssignID, AssignDate, AssignHours, ProjCode, EmpID)
```

Table 4

Write the SQL statements based on the following requirements:

a) Display the project names that were assigned from January 2025 until April 2025.

(4 marks)

b) Display the total charge for each employee and name the output as "Total Charge".

(5 marks)

c) Display the full names of the employees who have been assigned to more than two projects.

(5 marks)

d) By using a subquery, display the job name that has a charge per hour of RM50 and has more than 10 employees.

(6 marks)

QUESTION 5 (20 marks)

Bristledbeast is a hiking agency that manages various trips to hike local mountains. They plan to build a database system in order to keep track the organized trips throughout the years. You are hired to design and develop a database system based on the requirements needed are as following:

- Details of hikers need to be recorded in the system. Hiker identification number, full name, address, phone number, email.
- Each hiker also needs to give details of any family member. They need to give their family member's full name, emergency contact number, address and email.
- For each trip, the trip ID, trip name, approximate length (in hours), and fees charged are recorded in this table.
- Each hiker can join many trips. Start date of hike and end date of hike are also need to be recorded in this system.
- Each trip is managed by at least one guide. Some guides have been assigned to several trips or may not assign to any trip. Payment for each assigned trip is recorded in this system. Guides are identified by an employee ID. Guide full name, contact number, address, hire date are also recorded in this system.
- Each trip will have one destination of mountain. The details of mountain such as mountain name, height and country are also need to be stored in this system.

Based on the given information:

a) Construct an Entity Relationship Diagram (ERD) complete with all the required tables, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

b) Briefly explain TWO (2) examples of reports that can be generated from ERD in (a).

(4 marks)

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